

System Architecture: Publish/Subscribe Model

1. Event Metadata

- **Event Name / Topic:** `Package_Arrived_At_Hub`
- **Publisher:** **Scanner API** (Triggered by Barcode/RFID hardware at the warehouse entrance).

2. Subscriber Actions

The system uses a decoupled approach where multiple services listen for the arrival event and act independently:

Subscriber	Action Taken
Inventory Service	Updates the package status to "Arrived at Hub" in the central database.
Notification Service	Sends a Push/SMS alert to the customer confirming the package location.
Logistics Controller	Triggers the automated sorting system to route the package for the next leg of the journey.

3. Technical Flow

1. The **Scanner API** captures the package ID and publishes the `Package_Arrived_At_Hub` message to the **Event Broker** (Kafka/RabbitMQ).
2. The Broker broadcasts the message to all active **Subscribers**.
3. Each service processes the data concurrently without requiring direct communication with the Publisher.